

NGUYEN MANH DUNG

Backend Software Engineer

SUMMARY

Backend Software Engineer with 2+ years of experience building high-concurrency distributed systems using Node.js / TypeScript (NestJS) and AWS. Skilled in resolving concurrent processing and data consistency challenges, optimizing database performance, and designing resilient, event-driven microservices for production systems.

WORK EXPERIENCE

Ecosystem: Social Network & Dining Coupon Platform | Team size: 8 members

- Designed a 2-stage newsfeed ranking pipeline combining implicit feedback scoring and time-decay with cron-based trending updates, improving content relevance for personalized “For You” feeds.
- Resolved database bottlenecks for high-follower accounts by engineering a Hybrid Fan-out (Push/Pull) strategy, cutting newsfeed API response time from over 1s to **under 200ms**; optimized deep relationship queries by replacing complex SQL joins with Neo4j graph traversal.
- Improved maintainability of APIs by adopting CQRS with NestJS CommandBus (Mediator pattern), isolating business logic into dedicated command handlers to reduce service coupling.
- Prevented duplicate rewards and data corruption during high-concurrency flash-sale/gacha events by processing requests asynchronously with BullMQ and serializing claims via Redis Distributed Locks (Redlock), maintaining **zero data-loss** under traffic surges.
- Migrated a monolithic system into standalone microservices (Chat, Coupon, System), enabling independent scaling via gRPC for low-latency inter-service communication on AWS EC2.
- Prevented thundering-herd failures during **10,000+ notification broadcasts** by batching RabbitMQ consumers into groups of 500, keeping downstream CPU utilization stable under 40%.
- Maintained **90%+ Jest test coverage** for business-critical modules; delivered production releases via Bitbucket CI/CD pipelines, monitoring Docker containers and logs in staging under Agile/Scrum.

Cross-Platform Health & Fitness Platform | Team size: 2 members

- Implemented an Adapter Pattern to isolate core services from storage-provider SDKs, and built a Redis-based Proxy caching pipeline in front of core services, reducing direct database reads and latency during peak traffic.
- Reduced frontend network overhead by consolidating 4–5 REST API calls into a single GraphQL query for dashboard metrics, notification status, and related resources — eliminating redundant round-trips on page load.
- Replaced polling with Socket.io to push real-time updates for workout progress, heart rate, calories, step count, and notification badges.
- Reduced fitness video startup latency from ~800ms to **under 150ms** by setting up a video delivery pipeline on Amazon S3 and CloudFront CDN.
- Built an A/B testing pipeline to evaluate UI/UX experiments and workout template rollouts against real usage data, enabling the product team to make data-driven release decisions instead of relying on guesswork.

- Built scalable REST APIs using NestJS and TypeScript, implementing the Repository, Dependency Injection, and Template Method patterns to abstract base CRUD operations, decouple data access, and eliminate redundant boilerplate code.
- Streamlined frontend integration and pre-release testing by collaborating closely with Frontend and QA teams using Postman, accelerating bug identification and resolution cycles.
- Maintained high code quality and repository health by enforcing the Gitflow branching strategy and actively participating in peer code reviews.

TECHNICAL SKILLS

- Languages & Frameworks:** TypeScript, JavaScript, NestJS, NodeJS.
- APIs & Databases:** REST, GraphQL, gRPC, Socket.io, Firebase FCM, NoSQL & Caching (MongoDB, Redis), SQL (MariaDB), GraphDB (Neo4j).
- Queues & Cloud:** RabbitMQ, BullMQ, AWS (EC2, S3, CloudFront, IAM, Route53).
- DevOps & QA:** Docker, Nginx, Container Monitoring, Log Analysis, Jest, Mocking.
- Architecture & Tools:** Microservices, CQRS, Design Patterns, Git, Bitbucket, Jira, Slack, Agile / Scrum.

EDUCATION

Hong Bang International University (2021 – 2025)

Bachelor of Software Engineering

REFERENCES

Nguyen Vo Dang Cao — Product Manager, MetaGRIT Co., Ltd.